End-Term Sample Question Paper - SET C

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set C - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.

Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

Q1. State De Morgan's laws for sets, and write the complement of A ∪ B in words. **(2 Marks)**

[Unit I · CO1 · L1 - Remember · Theory]

Q2. Calculate the eigenvalues of A = [[3, 1], [1, 3]], and verify them with the trace and the determinant. **(2 Marks)**

[Unit II · CO2 · L2 - Understand · Numerical]

Q3. A server receives requests following a Poisson distribution with mean λ = 2 per minute. Compute P(X = 1), correct to four decimal places. **(2 Marks)**

[Unit III · CO3 · L2 - Understand · Numerical]

Q4. State the condition for a connected graph to have an Eulerian circuit, and the condition for it to have an Eulerian path but no circuit. **(2 Marks)**

[Unit IV · CO4 · L1 - Remember · Theory]

Q5. State what it means for a problem to be in P and in NP, and state what the Cook–Levin theorem established. **(2 Marks)**

[Unit V · CO5 · L2 - Understand · Theory]

Part B - Short Answer Questions (4 x 5 = 20 Marks)

Q6. Apply the 0/1 knapsack method to items of weights {3, 4, 5} and values {₹30, ₹50, ₹60} with capacity W = 8. List the feasible subsets, state the optimal value and the items chosen. **(5 Marks)**

[Unit VI · CO6 · L3 - Apply · Numerical]

Q7. In a survey of 120 students, 60 use Windows, 45 use Linux and 40 use macOS; 25 use Windows and Linux, 20 use Windows and macOS, 15 use Linux and macOS, and 10 use all three. Using the inclusion–exclusion principle, find how many use at least one of the three, and how many use none. **(5 Marks)**

[Unit I · CO1 · L3 - Apply · Numerical]

Q8. Distinguish a confidence interval from a hypothesis test, define Type I and Type II errors, and state the relationship between a 95 per cent interval and a two-tailed test at α = 0.05. **(5 Marks)**

[Unit III · CO3 · L2 - Understand · Theory]

Q9. Describe three methods for solving a recurrence relation — iteration, the recursion tree, and the characteristic equation — and state which kind of recurrence each suits. **(5 Marks)**

[Unit V · CO5 · L4 - Analyse · Theory]

Part C - Long Answer Questions (3 x 10 = 30 Marks)

Q10. Image transformation in homogeneous coordinates. **(10 Marks)**

[Unit II · CO2 · L4 - Analyse · Theory + Numerical]

(a) Explain why a translation cannot be written as a plain 2×2 matrix and how homogeneous coordinates solve this, and state why the order of composition matters. *[5]*

(b) A point is first translated by (3, 2) and then scaled by 2 about the origin. Write T and S as 3×3 matrices, compute M = S·T, and apply M to the point (1, 4), showing each stage. *[5]*

Q11. Graph colouring and timetabling. **(10 Marks)**

[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]

(a) Define the chromatic number, state the bound involving the size of the largest clique, and describe the Welsh–Powell heuristic. *[5]*

(b) Five papers A to E must be scheduled; the pairs sharing students are A–B, A–C, B–C, B–D, C–E and D–E. List the degrees, apply Welsh–Powell, and give the timetable with the number of slots. *[5]*

Q12. Divide and conquer, and solving its recurrences. **(10 Marks)**

[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]

(a) State the divide-and-conquer template, explain how Karatsuba and Strassen each reduce the number of sub-problems, and state why that lowers the exponent. *[5]*

(b) Solve: (i) $T(n) = 2T(n/2) + n^3$; (ii) $T(n) = 7T(n/2) + n^2$; (iii) $T(n) = 2T(n-1) + 3$ with $T(0) = 1$ — give the closed form and $T(6)$. *[5]*

----- End of Question Paper -----